

LAB GUIDELINES

Lab 8 — IP, VLAN, DNS, DHCP, FQDN, and Hosts File

Maps to CompTIA Server+ objective **2.2 — Given a scenario, configure servers to use network infrastructure services** (IP configuration, VLAN, default gateways, DNS, FQDN, hosts file, IPv4 RFC 1918, IPv6, firewall ports, DHCP, APIPA, MAC addresses).

Run all commands on <https://killercode.com/playgrounds/scenario/ubuntu>

Required software (free, apt): iproute2, vlan, bind9, dnsutils, isc-dhcp-server

```
apt update && apt install -y iproute2 vlan bind9 bind9utils dnsutils isc-dhcp-server
```

Step 1 — Inspect IP configuration (static vs. dynamic)

```
ip -c addr
ip -c route
ip link show | head
cat /etc/resolv.conf
```

Note the **MAC address** on each link/ether line and the **default gateway** (default via ...).

Step 2 — Configure a static IP and default gateway

```
ip addr add ip address of the the above/24 dev enp1s0
ip route add default via ip address dev enp1s0 || true
ip -c addr show enp1s0
```

Step 3 — VLAN tagging on a single NIC

```
modprobe 8021q
ip link add link eth0 name eth0.20 type vlan id 20
ip addr add 192.168.20.10/24 dev eth0.20
ip link set eth0.20 up
ip -d link show eth0.20
```

eth0.20 carries VLAN 20 tags. On a real network the switch port must be **trunked** with VLAN 20 allowed.

Step 4 — RFC 1918 private address spaces (memorise for the exam)

Prefix	Range	Default mask
10.0.0.0/8	10.0.0.0 – 10.255.255.255	/8
172.16.0.0/12	172.16.0.0 – 172.31.255.255	/12
192.168.0.0/16	192.168.0.0 – 192.168.255.255	/16

Plus: **169.254.0.0/16 = APIPA** — what a client self-assigns if DHCP fails.

Step 5 — IPv6 quick check

```
ip -6 addr
sysctl net.ipv6.conf.all.disable_ipv6
ping6 -c2 ::1
```

Every interface gets at least a **link-local fe80::/10** address derived from the MAC.

Step 6 — DNS server (BIND9) with a tiny zone

```
cat > /etc/bind/named.conf.local <<'EOF'
zone "lab.local" {
    type master;
    file "/etc/bind/db.lab.local";
};
```

EOF

```
cat > /etc/bind/db.lab.local <<'EOF'
$TTL 604800
@   IN  SOA ns1.lab.local. admin.lab.local. ( 2 604800 86400 2419200 604800 )
@   IN  NS  ns1.lab.local.
ns1 IN  A   127.0.0.1
web IN  A   10.99.99.20
db  IN  A   10.99.99.30
EOF
```

```
systemctl restart bind9
dig @127.0.0.1 web.lab.local +short      # → 10.99.99.20
dig @127.0.0.1 web.lab.local            # full record
```

web.lab.local is the **FQDN** = hostname + domain + trailing dot.

Step 7 — Hosts file (the local override that beats DNS)

```
echo "10.99.99.50 app.lab.local app" >> /etc/hosts
getent hosts app.lab.local
```

/etc/hosts (or C:\Windows\System32\drivers\etc\hosts) is consulted **before** DNS — the source of many "DNS issues" that aren't actually DNS.

Step 8 — DHCP server on a private subnet

```
cat > /etc/dhcp/dhcpd.conf <<'EOF'
default-lease-time 600;
max-lease-time 7200;
authoritative;
subnet 10.99.99.0 netmask 255.255.255.0 {
    range 10.99.99.100 10.99.99.200;
    option routers 10.99.99.1;
    option domain-name-servers 127.0.0.1;
    option domain-name "lab.local";
}
EOF
# Don't actually start the daemon on KillerCoda's shared net – just validate:
dhcpd -t -cf /etc/dhcp/dhcpd.conf && echo "Config OK"
```

The pool excludes .1 (gateway) and .2 – .99 (static reservations).

Step 9 — APIPA simulation

```
ip addr add 169.254.42.42/16 dev eth0
ip -c addr show eth0
```

If a Windows client cannot reach a DHCP server it self-assigns from **169.254.0.0/16**. Seeing a 169.254.x.x address is a tell-tale that DHCP is failing.

Free online tools

- ****IP Calculator**** — <https://alfredang.github.io/ipcalculator/>
- ****IANA IPv4 special-use registry**** — <https://www.iana.org/assignments/iana-ipv4-special-registry/>
- ****BIND9 ARM**** — <https://bind9.readthedocs.io/>

What you learned

- Static IP, default gateway, MAC address, and the difference between DHCP and APIPA (169.254.0.0/16).
- VLAN tagging on a single NIC with 802.1Q.
- RFC 1918 private ranges you must memorise for SK0-005.

- IPv4 vs. IPv6 quick checks.
- Building a BIND9 zone with A records and an FQDN.
- How `/etc/hosts` short-circuits DNS — and why that matters when troubleshooting.
- A minimal ISC DHCP server scope.